Ros Maria Edathil Francis Stanly

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EDUCATION

Master of Science - Computer Science **August 2024 - December 2026**
University of North Florida, Jacksonville, Florida **3.88 GPA**
Relevant Course Work: Advanced Artificial Intelligence, Design & Analysis of Algorithms, Programming II (Java), Database, Data Structures, UI/UX Design

Master of Technology - Electronics and Communication Engineering **July 2019 - June 2021**
Cochin University of Science and Technology, Kochi, India **4 GPA**
Course Work: Design Verification and Testing, Advanced Digital System Design, Advanced Signal Processing, VLSI Design and Verification, Advanced Embedded System Design

Bachelor of Technology - Electronics and Communication Engineering **July 2014 - June 2018**
Mahatma Gandhi University, Kottayam, Kerala **3 GPA**
Course Work: Programming I (C#/C++), VLSI Design, Computer Networks and Architecture, Communication Systems, Signal Processing.

PROFESSIONAL EXPERIENCE

VVDN Technologies Private Limited **Nov 2021 - Aug 2024**
Digital Signal Processing (DSP) Engineer *Kochi, India*

- Developed and optimized DSP algorithms for wireless 5G communication systems on CPU, GPU, and FPGA platforms.
- Extracted highly accurate timing and spatial data from wireless signals in multipath and complex environments.
- Designed algorithms in Python for signal extraction, visualization, and performance analysis.
- Performed fixed-point and fractional-point conversions between MATLAB, C, and Verilog RTL, ensuring precision and hardware efficiency.
- Built test setups (Vector Signal Generator , Vector Signal Analyzer), simulated and debugged algorithms, and tuned system performance in lab environments.
- Analyzed and processed signal data in MATLAB and Python, enhancing hardware performance and accuracy.
- Collaborate with design and verification engineers to analyze and resolve RTL and gate level simulation issues.

VVDN Technologies Private Limited **Aug 2021 - Oct 2021**
Digital Signal Processing (DSP) Engineer - Intern *Kochi, India*

- Implemented 5G channel algorithms in MATLAB and Python.
- Performed fixed-point and fractional-point algorithm conversions to ensure hardware efficiency.
- Documented algorithms, code implementations, and test results for team reference and validation.

TECHNICAL SKILLS

Language & Frameworks: Python,Java,C,SQL,TensorFlow, PyTorch, Scikit-learn, Keras, XGBoost,Hugging Face, Librosa.
Tools & Technology: Jupyter Notebook, Google Colab, Tableau, MySQL, AWS, Github, Linux.

PROJECTS

Multimodal Retrieval-Augmented Generation for Academic Resource Navigation – Python | PyTorch | CLIP| Whisper | NetworkX | PyTorch Geometric | LLaMA | Mistral |

- Designed and implemented a multimodal RAG system for information retrieval from PDF, slides, videos, images, and audio resources.
- Extracted and processed multimodal data with the help of libraries like PyMuPDF, python-pptx, OpenCV, and Whisper.
- Created a heterogeneous knowledge graph and performed semantic re-ranking of the results with the help of GNN (GCN and GAT)
- Integrated locally hosted LLMs like LLaMA and Mistral for answer generation.
- Performed the evaluation of the retrieval system with the help of Recall@K, MRR, and NDCG.
- Performed the evaluation of the generation effectiveness using BLEU, ROUGE-L, BERTScore.

Software Requirements Clustering and Labeling using Large Language Models – Python | NLP | Sentence Transformers | scikit-learn | LLaMA | Mistral |

- Designed a semantic enrichment pipeline based on Large Language Models to normalize and structure unstructured natural language software requirements for clustering.
- Used Sentence-BERT to create semantic embeddings of unstructured requirements, then applied several clustering algorithms such as k-means, hierarchical, and HDBSCAN to cluster requirements.
- Designed an automated system to label requirement clusters using Large Language Models.
- Used metrics like ARI, and NMI to measure the quality of requirement clusters, as well as PCA and UMAP visualization to measure separability between clusters.

Machine learning attack on hybrid current mirror inverter physical unclonable function – Google colab | Python | Librosa | Scikit Learn | Numpy |

- Designed and simulated a hybrid current mirror inverter based physical unclonable function using HSPICE.
- Generated the dataset through circuit level simulation and collected 100,000 challenge response pairs under modeled process variations to reflect realistic behavior.
- Built a machine learning evaluation pipeline in Python and trained and compared ML models including SVM, Logistic Regression, Decision Trees, and KNN.
- Achieved the best attack performance using an SVM classifier with a prediction accuracy of 92.55%.
- Performed PCA and UMAP based visualization to analyze feature separability and model behavior.

Voice Based Phishing Detection Using Machine Learning and Deep Neural Networks - Python | Google Colab| Librosa| Scikit Learn |TensorFlow | Signal Processing |

- Designed an end to end signal level voice phishing detection system resilient to noisy and telephonic environments using Kaggle Phishing Voice Dataset .
- Incorporated audio preprocessing steps for waveform scaling and noise consideration and derived handcrafted spectral and cepstral features like MFCCs, log-Mel spectrograms, and temporal features.
- Trained and assessed classical machine learning models like Logistic Regression, SVM, and Gradient Boosting for classification based on MFCCs.
- Implemented a CNN for a log-Mel spectrogram representation based learning system for distinctive spectral temporal regions.
- Recorded a maximum F1-score of 0.91 for CNN, outperforming other machine learning models used for comparison.
- Applied Grad-CAM visualization to interpret model decisions and highlight phishing specific spectral regions.

Hybrid Sampling-Bug Algorithm for Path Planning in Unknown Environment - MATLAB | RRT ToolBox |

- Developed a hybrid path planning algorithm that utilizes both Rapidly Exploring Random Tree (RRT) with RT-RRT sampling and Tangent Bug obstacle boundary following.
- Implemented the algorithm in MATLAB, integrating sampling-based global exploration with reactive local obstacle avoidance.
- Evaluated performance across 6 simulated environments with varying obstacle densities, measuring path length, execution time, and navigation success rate.
- Achieved 100% navigation success rate with path length and computation time comparable to that of RT-RRT.

Fast and Flexible Sampling-Based Local Replanning for Single Query Paths in Unknown Environments - MATLAB | RRT ToolBox |

- Designed a novel path planning algorithm combining RRT* for asymptotic optimality with local replanning upon collision detection.
- Implemented the algorithm in MATLAB using the RRT Toolkit for simulation and evaluation and designed local replanning mechanisms to dynamically repair paths instead of restarting global planning.
- Evaluated algorithm performance using path length, planning time,average success rate across 10 independent runs.
- Demonstrated improved timing efficiency and 100% success rate compared to standard RRT* and RRTX based replanning.

PUBLICATIONS

- Fast and Flexible Sampling-Based Local Replanning for Single-Query Paths in Unknown Environments - 39th International FLAIRS Conference.

CERTIFICATIONS AND ADDITIONAL INFORMATIONS

- Won the Best Computer Advisory Board Choice Award for poster presentation at the School of Computing Symposium on December 4, 2025 - Voice Based Phishing Detection Using Machine Learning and Deep Neural Networks.
- Developed city level and census tract level affordability indices using housing, transportation, walkability, and income data as part of the AI for Good Hackathon 2025.
- Certificate in embedded systems and hardware design – SMEC Labs
- The complete python bootcamp from zero to hero in python – Udemy
- Hands-on Workshops on IC design – Department of Electronics, CUSAT
- Digital Design Hands On - Maven Silicon
- Ready for Research - University of North Florida